WPI INTERACTIVE QUALIFYING PROJECT

#TeamGraduate AI Pedagogical Guidelines

A SET OF CONTENT CREATION GUIDELINES FOR #TEAMGRADUTE'S AI TOOLS

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1 Introduction

To ensure that the AI-powered tools developed by the project team for the #TeamGraduate platform produce consistently high-quality content that is effective at targeting the learning outcomes outlined by the Namibian secondary school curriculum, it was decided that the AI must have some form of guidance in creating each of the different content types.

Thus, a literature review was conducted that investigated the research behind the pedagogical best practices that should be employed when creating educational content. This included an initial investigation of what educational frameworks are most effective, and should thus be considered in the design of all content for the #TeamGraduate platform. Subsequently, more research was performed to create a set of guidelines for each specific type of AI-generatable content that the project team implemented into the #TeamGraduate platform.

2 Content Generation Guidelines

2.1 Educational Frameworks

When creating or generating any educational content, there must be some thought put into the framework behind the design of this content. Specifically in the context of education, there exist many well-established, well-researched frameworks to consider.

The most prevalent of these educational frameworks is blocked practice. This framework is defined as the assignment of many similar problems in immediate succession. It is widely used, particularly in mathematics, despite a lack of empirical evidence that supports its effectiveness for long-term learning (Rohrer et al., 2020).

A more effective alternative is spaced practice, defined as the distribution of practice for any given concept across multiple sessions separated by a period of time. Unlike blocked practice, this approach to education has received much support as an effective method for long-term education across decades of research (Cepeda et al., 2006), and its benefits are not uniform over time: Cepeda et al. (2008) found that spacing produced no consistent improvement in test scores after short delays. However, after delays of four weeks or more, spacing approximately doubled scores, suggesting that the advantages of spacing compound over time.

Therefore, generated content that covers any given topic should consider this and distribute topics across multiple assignments, rather than concentrating them within a single session.

A closely related principle, interleaved practice, addresses a further limitation of blocked formats. In a blocked assignment, students can know which strategy is required before reading the problem, as the assignment structure provides this information. This removes the inferential demands that are posed by assignments and real-world contexts (Rohrer et al., 2020). Interleaving mixes problem types drawn from different topics within a single assignment. This blending of content types has been shown to improve learning performance by approximately 30% over blocked conditions (Rohrer & Taylor, 2007), with no condition identified in which blocking produces superior outcomes (Rohrer et al., 2020).

For subjects based on problem-solving, mixing problems from previously taught topics with current content on the same assignment satisfies both the requirements of spacing and interleaving simultaneously. However, it should be noted that interleaving is most effective when applied to related problem types. Mixing entirely unrelated content within a single session is not supported by any existing empirical data, and may be counterproductive (Kornell & Bjork, 2008).

As a final note, it must also be considered that students themselves may misjudge the relative effectiveness of different practice frameworks. Interleaved and spaced practice are typically perceived as more difficult and less enjoyable than blocked practice, which may lead students to prefer the less effective format (Kornell & Bjork, 2008). This suggests that the platform's design must not be driven by learner preference alone, and that, where appropriate, framing should accompany practice activities to help students understand the reasons for using a mixed practice structure.

The design of all AI-generated content for #Team-

Graduate's platform is grounded in these two frameworks. Beyond these frameworks, to further ensure that all educational content generated for the #TeamGraduate platform is as effective as possible, consideration was put into the learners' context. Any content generated will be more effective if it is locally relevant and engaging to the target population. Furthermore, all content generated should be hyper-specific to the learning outcomes of the curriculum to be taught, which will ensure that all content generated does not waste the time or energy of the learners covering unnecessary content.

2.2 Practice Problems

The design of practice problems within the #TeamGraduate platform is grounded in two principles: spaced practice and interleaved practice. Together, these form the basis for decisions about problem sequencing, variety, and feedback.

Problem Design Within individual problem sets, problems should progressively increase in difficulty, with Rosenshine (2012) identifying an approximately 80% success rate as the optimal threshold. Around this success rate, there is a sufficient blend of learning and cognitive demand that facilitates meaningful practice. Sets should include both routine problems, which build fluency in problem-solving procedures, and non-routine problems, which require students to identify an appropriate strategy from several possibilities.

Non-routine problems are particularly valuable in interleaved problem sets, where strategy selection is the central cognitive challenge that blocked practice eliminates (Rohrer & Taylor, 2007).

Feedback To better help with information retention and material fluency, practice sets should be accompanied by worked solutions that explain the reasoning behind each step in the problem-solving process. This is due to the fact that feedback that provides diagnostic information, such as problem-solving logic, has been shown to substantially outperform feedback that communicates only a grade (Morris et al., 2021).

However, students should be encouraged to attempt all problems independently before consulting such worked solutions, as passive review of worked examples without prior effort to attempt such problems reduces the learning benefit that feedback provides (Rosenshine, 2012).

Application to English Language Practices While most of the above content discusses problem creation logic in relation to mathematics, all the principles theoretically extend to English language practices. English content should also adhere to the frameworks of spaced and interleaved practice; vocabulary and grammar topics should be distributed across multiple sessions rather than introduced and abandoned. Furthermore, practice sections should mix task types such as grammar, comprehension, vocabulary, and writing rather than adopting a blocked practice framework which would individually group each of these topics (Cepeda et al., 2006).

It should be acknowledged that while generally these principles and frameworks can, in theory, extend to applications in the English language, there is much less empirical evidence to support their validity in English than in mathematics, and as such this application represents a theoretically motivated extension of the larger framework rather than an independently evidenced finding (Kornell & Bjork, 2008).

Furthermore, where possible, practice problems should include a mixture of both recognition tasks, where the learner must identify previously learned information when it is presented, and production tasks, which require the learner to actively generate information without prompts. Including both types of tasks introduces the benefit of a variation in difficulty due to the fact that production tasks impose greater cognitive demand and are associated with stronger learning outcomes (Rosenshine, 2012).

What to Avoid Several common practices in creating problems are inconsistent with the evidence outlined above and should be avoided in generating any content for #TeamGraduate's platform.

1. Practice questions should avoid the blocked practices framework, and should thus not be grouped by content type or theme, nor should they be generated without any deliberate variation in format, difficulty, and context (Rohrer et al., 2020).

2. Sets of practice questions should not be assigned prior to adequate instruction on the covered content (Rosenshine, 2012).

Specific Question Type Guidelines

1. **Fill-in-the-Blank**

   Fill-in-the-blank questions should be utilized primarily for facilitation of active recall of specific terminology and facts, rather than as higher-order assessments. Effective question construction requires that the missing information is central to the learning objective, as removing trivial words can distract learners from the core concepts (Jenner, 2024).

   Furthermore, as a retrieval-based activity, these tasks should only follow initial instruction, serving to strengthen existing knowledge, rather than introduce new concepts (Karpicke, 2012).

   Fill-in-the-blank questions should always be generated in a manner that ensures that the context surrounding the blank is sufficient to lead to a single, objective answer to minimize ambiguity. Furthermore, as previously stated, fill-in-the-blank questions are better served as a tool for surface-level evaluation rather than for evaluating complex reasoning. This is because this format may present linguistic barriers for learners who struggle with spelling or specific vocabulary, which could potentially prohibit learners from actually understanding the content (Jenner, 2024).

   Finally, to maximize the effectiveness of fill-in-the-blank questions, immediate feedback must be provided to correct errors and reinforce the retrieval of accurate information.

2. **Drag-and-Drop**

   Drag-and-drop questions are most effective for tasks that involve categorization, labeling, and sequencing, where the physical manipulation of elements can reinforce the learning of spatial and conceptual relationships. Drag-and-drop questions are particularly useful as the tactile act of "sorting" can help make abstract structures more concrete by creating physical things out of mental operations (Lehrman, 2025).

   However, these questions should be designed following the spatial contiguity principle, which requires that feedback and instructions are placed in close proximity to the interaction area to reduce unnecessary cognitive load (Mayer, 2009). Designers must provide immediate, task-level feedback that explains the logic behind correct and incorrect placements, as this type of information is critical in modifying a learner's behavior in real time (Shute, 2008).

   Similar to fill-in-the-blank questions, drag-and-drop items should not be utilized as the primary method for assessing complex reasoning, as the simple act of matching can facilitate guesswork, rather than genuine comprehension.

3. **Spot-the-Error**

   Spot-the-error questions require that learners identify deliberate mistakes within a worked solution or passage, explain the underlying misconception, and produce a corrected version. Similarly to fill-in-the-blank and drag-and-drop questions, this format is most effective when implemented after initial instruction, as it requires students to test their understanding against flawed material (Rushton, 2018).

   Spot-the-error questions are effective as they force learners to investigate the logic of a solution rather than just passively reviewing worked examples, which significantly improves long-term information retention and depth of material fluency (Dieterich et al., 2025).

   However, just like other question formats, spot-the-error questions are most effective when implemented in a specific manner. Research indicates that simply identifying an error pro-

duces weaker learning outcomes than tasks that require written justification (Dieterich et al., 2025). Thus, spot-the-error questions must target common, genuine misconceptions and require students to articulate why a specific step is incorrect. To maximize learning opportunities, errors should be embedded early in a multi-step process, as this allows students to trace the consequences of these errors.

To be consistent with #TeamGraduate's platforms feedback framework, these questions must provide immediate feedback that explains the source of the error and the reasoning behind the correct approach.

2.3 Flashcards

The creation of flashcards within the #TeamGraduate platform is grounded in the principle of active retrieval, which can be defined as the process of calling information to mind from memory. Unlike passive review, the act of retrieval is not merely a metric of learning, but is itself a method of strengthening neural pathways that facilitate long-term information storage (Karpicke & Roediger, 2008).

Card Architecture and Cognitive Load To ensure that flashcards are as effective as possible, they must adhere to an "atomic" design, where each card targets a single concept or relationship. This structure ensures that the learner's cognitive effort is focused entirely on the target retrieval, facilitating the successful retrieval necessary for long-term information retention (Karpicke & Roediger, 2008).

Furthermore, when appropriate, the design of flashcards should integrate concise verbal prompts with relevant visual aids. By providing both a linguistic and a graphic representation of a concept, the flashcards create multiple, independent memory traces. This redundancy increases the probability of successful recall during future assessments (Mayer, 2009).

Scheduling and Retention Strategy Flashcards are a means of learning that must be strictly adherent to the previously discussed spaced repetition framework. Empirical research demonstrates that spacing retrieval attempts across increasing intervals of time is significantly more effective the blocked practice, even when the total time spent studying is identical (Kornell, 2009).

What to Avoid Content generated for flashcards must avoid several specific design flaws that stand to undermine the retrieval process.

1. **Contextual Cues** Cards should not contain any hints or patterns that allow the learner to guess the correct answer through recognition rather than active recall.

2. **Complex Syntax** All cards generated, specifically those used for English language content, should utilize simple, easily understandable jargon. It is imperative that the cognitive effort of the learner not be wasted in attempting to understand the question.

3. **Premature Retirement** All flashcards generated should not be hidden or removed from the rotation after a single successful answer. Sustained material fluency requires repeated cycles of retrieval over time (Kornell, 2009).

2.4 Assessments

All assessments generated within the #TeamGraduate platform are designed with deliberate question construction that adheres to the necessary learning outcomes. These design choices form the basis for decisions about question type selection, cognitive demand, and the approach of the platform towards AI-generated content.

Purpose and Question Design Before generating any assessment, quiz, or exam, its purpose must be determined. This requires considering whether it should be a formative assessment, where ongoing feedback is provided during learning, or a summative assessment, which evaluates the degree of mastery of learning outcomes at the end of an instructional period. Every question must adhere strictly to a specific learning outcome and cover only a single topic. Across a whole assessment, questions

should vary in difficulty and format, and the question types selected should match the nature of what is being assessed.

Multiple-choice questions (MCQs) are the most versatile format for assessing factual and conceptual knowledge when they are well constructed. Effective MCQ stems should contain the full central idea of the question, such that a student could anticipate the answer before reading any of the options. Incorrect answers should distract learners with genuine misconceptions rather than simply unlikely alternatives, and all MCQ answer options should be kept consistent in length and format to prevent confusion (May et al., 2025).

Bloom's Taxonomy Alignment Throughout an assessment, questions should be distributed across difficulty levels that are consistent with Bloom's revised taxonomy: a framework that categorizes educational objectives and goals into the six bins of increasing cognitive difficulty shown below:

1. Remembering
2. Understanding
3. Applying
4. Analyzing
5. Evaluating
6. Creating

Lower-order questions at the Remember and Understand levels should be included as they are appropriate for assessing foundational knowledge. These types of questions should use clear, direct language with plausible, distracting answers that are grounded in common errors. Higher-order questions at the Apply, Analyze, and Evaluate levels should also be included as they require learners to apply reason to eliminate distracting answers that are grounded in incomplete or incorrect reasoning (May et al., 2025).

Assessments should contain a variety of questions and should avoid concentrating the difficulty level of these questions at the lower cognitive levels. Over-reliance on lower-order questions limited the diagnostic value of an assessment and can fail to distinguish between surface-level familiarity and genuine content mastery (Law et al., 2025).

Limits of AI-Generated Content For the #TeamGraduate platform, because all assessments and other content can be generated using AI, it is important to consider the limitations of AI-generated content. May et al. (2025) found that 80% of MCQ items produced by AI-based question generators violated at least one of the guidelines for creating effective MCQs. The most prevalent violations to be cautious of were unequal answer lengths, the existence of multiple defensible answers or no correct answer, unlikely distractor answers, and the use of unnecessarily complex phrasing in question stems and answers. Nearly three-quarters of items in the study were assessed as likely to produce a major measurement error (May et al., 2025).

Furthermore, AI-generated questions tend to be of lower cognitive levels. Law et al. (2025) found that AI-generated MCQs assessed lower-order cognitive skills at a much higher rate than human-created questions. AI tools can also be prone to hallucination, which can cause questions to contain incorrect content.

Human Review Given the limitations of AI-generated content, all assessment items created with generative AI (GAI) should undergo some level of human review. Before being used, May et al. (2025) recommends that all content created using GAI should be checked against the following "content validity checklist":

1. Is the content aligned with what was actually taught?
2. Is the item keyed correctly?
3. Is there one and only one correct answer?
4. Are all response options logical and optional?

When a piece of AI-generated content does not meet each of these criteria, it must be revised prior to use. This is why it is important to use AI-generated content, particularly MCQs, as a starting point and not as a finished product (May et al., 2025). Furthermore, teachers should never blindly

trust AI outputs for assessment materials, as errors in generated content could harm the validity of the student performance analysis.

What to Avoid Several common practices are best avoided when creating assessment materials:

1. Use of negative constructions in questions, such as "which of the following is NOT..." or "all of the following EXCEPT ...", as these promote logical elimination rather than knowledge retrieval (May et al., 2025).

2. Use of absolute terminology such as "always" or "never" in distractor answers, as these signal incorrectness and should be avoided.

3. Use of catch-all terminology such as "all of the above" or "none of the above", as these introduce guessing advantages.

2.5 Study Guides

Study guides on the #TeamGraduate website are designed as tools that integrate retrieval style practice with cognitive load management.

Structure and Content A study guide is made to serve as a tool that breaks complex material into more manageable components, thus directing student attention toward the most important content. Every study guide created should be aligned directly with specific learning outcomes, and any content outside these learning outcomes should be excluded. The goal of a study guide should be deliberate focus rather than complete comprehensiveness.

All study guides on the #TeamGraduate platform should be consistent with cognitive load theory. This implies that information should be carefully fragmented and presented as singular concepts, as working memory has a limited capacity to process multiple streams of new material simultaneously (Sweller, 1994).

This means that, ideally, content should be sorted into clear sections, with logical headings, any text should be kept concise, and visual elements should be used in place of dense text blocks where appropriate.

Retrieval Practice Integration Effective study guides should not be purely passive reading materials. Rather, each section should include self-testing elements such as practice questions, reflection activities, or recall practice that requires students to retrieve key ideas without reviewing the content. This idea is grounded in a large body of evidence that shows that actively recalling information from memory significantly improves long-term retention compared to re-reading the same material (Roediger & Karpicke, 2006).

Furthermore, to adhere to spaced learning practices, study guides should be designed to be reviewed in multiple spaced sessions, rather than being read once in full. Therefore, the structure and section length of the AI-generated study guides should reflect this kind of repeated review style rather than encouraging passive consumption.

What to Avoid Several common design choices may undermine the effectiveness of study guides, and should thus be avoided in their creation within the #TeamGraduate platform:

1. Study guides should not attempt to review all material comprehensively, and material not tied directly to the addressed learning outcomes should be excluded as it adds cognitive load without increasing learning value (Sweller, 1994).

2. Study guides should not exist as continuous blocks of text without any visual breaks or headings.

3. Study guides should not be assigned as a single-use activity, as without repeated practice across spaced intervals, the benefit of study guides is substantially decreased (Roediger & Karpicke, 2006).

2.6 Diagrams

Two principles guide the design of diagrams within the #TeamGraduate platform: dual coding theory and cognitive load management. These principles form the basis for decisions about when to use diagrams, what information they should contain, and

how they can be effectively implemented into neighboring text.

Purpose and Visual Design Diagrams are most effective when they communicate an idea that text alone cannot effectively convey, such as spatial relationships, sequential processes, comparisons, and conceptual hierarchies.

According to dual coding theory, information encoded through both visual and text channels produces a stronger memorization effect than either channel alone (Clark & Paivio, 1991). This implies that diagrams are best when integrated with written explanations rather than as direct substitutes for text.

However, the benefit of dual coding is contingent on the quality of the integration. Mayer (2009) established a cognitive theory of multimedia learning that states that visual and text-based information must be aligned conceptually and spatially. Furthermore, labels, explanations, and any associated text should appear close to relevant portions of the diagram (Mayer, 2009). This is because when related information is split up, students must use cognitive effort to integrate the information, which produces unnecessary cognitive load.

What to Avoid Several common practices must be avoided when generating diagrams for the #TeamGraduate platform, as they can make their implementation less effective:

1. Diagrams should not be used as a decoration, as any diagram that does not represent a structure that genuinely benefits from visual encoding adds visual clutter that competes for attention (Mayer, 2009)

2. Diagrams should not contain too much information, as overwhelming learners with too much information in one visual component can create additional cognitive strain on top of processing the material.

3. Where complex information must be represented, it should be split up across multiple simpler diagrams.

2.7 Worksheets

Worksheets on the #TeamGraduate platform are designed as active learning tools, structured to build cognitive complexity progressively across each task.

Purpose and Question Design Worksheets are most effective when they serve as structured learning activities that require learners to construct an understanding rather than passively receiving it. Research found that worksheets produced measurable gains in conceptual understanding and improved the ability of students to connect them with real-world application (Buffalari, 2022).

Worksheets should be aligned with specific learning outcomes, and all questions should be sequenced in order of increasing cognitive demand, starting with recall, and progressing towards application and analysis.

Furthermore, questions should mirror the types of tasks that students will encounter in assessments, and where possible, task types should be labeled explicitly by cognitive level. Such transparency can help reduce exam anxiety among learners and support metacognition by helping students to identify the strong points of their knowledge, and where their understanding would benefit from further preparation (Buffalari, 2022).

Design Principles Worksheets, like many other content types, should be designed to minimize cognitive load. Directions should be short and unambiguous, text should be kept minimal, and each worksheet should ideally focus on a single task type. Where multiple task formats must be included, a clear visual distinction between sections is essential in preventing confusion.

What to Avoid As for all content types, some common practices should be avoided when generating worksheets, as they can potentially hinder their benefits:

1. Worksheets should not be graded for accuracy when used as practice tools.

2. Worksheets should not contain ambiguous instructions, insufficient space for student re-

sponses, or unnecessary design elements, as each of these represents an avoidable source of cognitive load (Buffalari, 2022).

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2.8 Reading Comprehension Passages

Reading comprehension passages for the #TeamGraduate platform are designed specifically to provide strategy-based instruction and a progressive difficulty experience that involves questions of varying cognitive levels.

Text Selection and Question Design Reading comprehension is a cognitive process that requires active learning by requiring students to construct meaning through the integration of their knowledge and experiences, rather than simply requiring the passive extraction of meaning from text (McNamara & Magliano, 2009). Therefore, passages should be selected at an appropriate level of vocabulary, as excessive unfamiliar vocabulary consumes cognitive resources that could otherwise be put towards creating meaning.

Texts should ideally be varied in formats, which could include short stories, essays, and informational paragraphs, and should be relevant to the reader's real-world contexts. This is a particularly important consideration for the AI-generated content on the #TeamGraduate platform, as all generated content should be locally relevant for Namibian learners.

Questions should once more be distributed across cognitive levels and structured such that they progress from lower-order to higher-order thinking. Questions that require inference are also particularly important, as research has shown that students' ability to generate inferences by combining textual evidence with prior knowledge to reach conclusions that are not stated explicitly in the text is strongly associated with overall reading comprehension outcomes (Elleman, 2017).

Scaffolding Comprehension When possible, reading comprehension tasks should engage students' prior knowledge before the reading passage, as this existing knowledge base substantially affects their ability to draw inferences, retain information, and construct accurate meaning from text (McNamara & Magliano, 2009). During and after reading, questions must require that students return to the text rather than relying solely on memory recall. Furthermore, tasks should require both recognition-based and production-based responses to provide a variance in cognitive difficulty.

What to Avoid The following common practices used in creating reading comprehension tasks should be avoided in the content generated for the #TeamGraduate platform, as they stand to undermine the educational value of comprehension passages:

1. Passages that are too vocabulary-heavy and don't have sufficient contextual support should be avoided. So too should sets of accompanying questions that focus too heavily on active recall (Elleman, 2017).

2. Instruction that uses comprehension strategy should not be introduced before students are aptly prepared. Students who must use significant cognitive effort to recognize certain vocabulary will have reduced working memory capacity to be used for actual comprehension of the passage.

3 Conclusion

From these guidelines, it can be concluded that there are several overarching themes that should be heavily considered in the creation of any educational content.

Generally, to best promote effective learning, content should follow the spaced and interleaved practice frameworks. This requires that content be designed intentionally so that it encourages practice across multiple spaced-out sessions, and also that concepts on an assignment are varied in type and content.

Furthermore, content should be designed with Bloom's taxonomy in mind. This means that in creating educational content, there is an obligation

to consider the cognitive levels of different questions, and include a variety of these question types in order of increasing cognitive difficulty.

A further common theme that emerged was that almost all content types that the #TeamGraduate platform is capable of generating are most effective when assigned after adequate instruction on the covered material. Thus these resources should be used only after a student has enough existing knowledge to make reviewing content beneficial.

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